MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER
TYPES 2N4865, 2N5250, 2N5251 and
TX2N4865, TX2N5250, TX2N5251

This interim amendment is issued for use by Rome Air Development Center, Department of the Air Force, with Military Specification MIL-S-19500/380A(EL), dated 17 September 1968.

PAGE 1

TITLE, delete and substitute:

"SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER
TYPES 2N4865, 2N5250, AND 2N5251
NON-TX, TX, AND TXV"

1.1: At end of paragraph, add "The prefix 'TXV' is used on devices submitted to and passing the internal visual inspection specified in 4.5."

PAGE 2

1.4, h_{FE} at I_C = 20Adc: Under Max column, delete "120" and substitute "90" (three places).

PAGE 4

Add the following new paragraphs:

"3.3.2 Internal visual (PRECAP) inspection and process-conditioning, testing, and screening of 'TXV' types. The 'TXV' device type shall, in addition to all performance requirements, be internally visually inspected and process-conditioned, tested, and screened in accordance with 4.5."

"3.4.2 'TXV' marking. Devices in accordance with the 'TXV' requirements shall be marked with 'TXV' immediately following the 'JAN' prefix."

PAGE 5

4.2.1, line 2: Delete "TX" and substitute "TX and TXV". Make this same change in line 5.

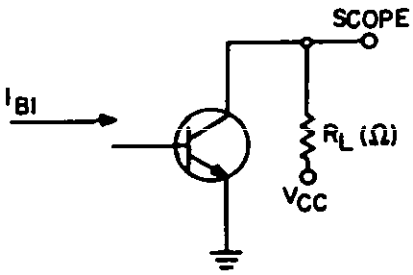
PAGE 11

Add the following new paragraph:

"4.5 Internal visual (PRECAP) inspection and process-conditioning, testing, and screening of 'TXV' types. The internal visual inspection shall be performed in accordance with test method 2072 of MIL-STD-750 prior to encapsulation on a 100 percent basis and process-conditioning, testing, and screening shall be as specified in 4.4. The manufacturer shall permit the authorized government representative to witness concurrent with time of manufacturer's performance of these tests, the process-conditioning, testing, and screening of the devices. Those conditioning and screening tests normally performed by a manufacturer as standard production tests, need not be repeated when these are predesignated and acceptable to the Government as being equal to or more severe than the test specified herein."

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Ⓐ at: I_C = 70 Adc



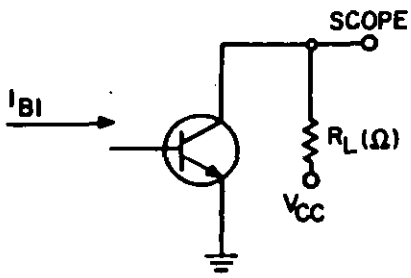
CIRCUIT CONDITIONS:
$10 \leq V_{CC} \leq 25$ Vdc

R_L (non-induc.) = $\frac{V_{CC} - V_{CE}(sat)}{70}$

where: $V_{CE}(sat)$ = value at I_C = 70A, I_B = 10A

$I_{B1} = -I_{B2}$ = 10A

Ⓑ at: I_C = 40 Adc



CIRCUIT CONDITIONS:
$10 \leq V_{CC} \leq 25$ Vdc

R_L (non-induc.) = $\frac{V_{CC} - V_{CE}(sat)}{40}$

where: $V_{CE}(sat)$ = value at I_C = 40A, I_B = 4A

$I_{B1} = -I_{B2}$ = 4A

FIGURE 3. Pulse response test circuit(s).

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PAGE 12

Table I, LTPD column: In TX subcolumn, add "TXV". Make this same change on pages 13 through 16 (table I); pages 17 through 24 (table II); and pages 25 and 26 (table III).

PAGE 13

Table I, Subgroup 3, h_{FE} (first test), Max Limits: Delete "120" and substitute "90".

PAGE 27

Figure 1, VIEW A: Delete "$\frac{.510}{.550}$" and "$\frac{.460}{.420}$" and substitute "$\frac{.470}{.550}$" and "$\frac{.380}{.460}$", respectively.

PAGE 29

Figure 3: Delete in its entirety and substitute new figure 3 as printed on page 2 of this amendment.

PAGE 32

Figure 7: Delete "R_S = 1Ω" and substitute "R_S = .1Ω"; also delete clamp voltages in the testing procedure and substitute the following:

"2N4865 = 80V
2N5250 = 100V
2N5251 = 150V"

Custodian:
Air Force - 17

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5961-F702)

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